Review II

1. Section 2.3
    a. Algebra of Limits
    b. Limits of Polynomials, Rational Functions, Trig Functions, Exp/Log Functions
    c. Limits of Piece-wise Functions

    d. SP: 5, 12, 16, 17, 21, 27, 42, 54

2. Section 2.4
    a. Continuity at a Point
        i. Intuitive Notion
        ii. Limit Definition
    b. Theorem 2.3
    c. Algebra of Continuous Functions
    d. Continuity on an Interval
        i. Open Interval
        ii. Closed Interval
    e. Intermediate Value Theorem
        i. Zero (Root) of a Function

    f. SP: 8, 11, 14, 19, 26, 32, 36

3. Section 3.1
    a. Slope of Tangent Line = Limit of Slopes of Approximating Secant Lines
    b. Limit Definition of Derivative
    c. Equation of Tangent Line to the Graph of a Function
        i. Equation of Normal Line to the Graph of a Function
    d. Relation of the Graph of the Derivative of a Function to the Graph of the Function
    e. Derivative Notation

    f. SP: 21, 23, 29, 33, 37

4. Section 3.2
    a. Basic Rule for Differentiating
    b. Differentiation of Algebraic Functions
    c. Linearity of Derivative
    d. Higher Derivatives

    e. SP: 11, 13, 15, 17, 19, 22, 28, 29, 33, 35, 43

5. Section 3.3
    a. Rules for Differentiating Transcendental Functions

    b. SP: 3, 7, 9, 12, 17, 23, 26, 32, 35, 39, 41, 49, 53

6. Section 3.4
    a. Instantaneous Rate of Change = Limit of Average Rate of Change
    b. Linear Motion
        i. Position, Velocity, Acceration, Speed, Total Distance
    c. Falling Body Problem

    d. SP: 3, 5, 9, 16, 18, 24, 29, 36

7. Section 3.5
    a. Chain Rule

    b. SP: 17, 20, 21, 24, 26, 31, 34, 37, 40, 41